

EUSOPHY — The Supreme Intellectual Endeavor

This Treatise is an evolved version of its **Japanese counterpart** — far beyond a mere translation — and is progressing toward its own completed form. It expands the non-mathematical portions of my paper "**Foundations of Eusophy** — A mathematical concept of God for the reconciliation of religion and science" (hereafter FE). FE itself grew out of my earlier monograph "**Mathematical Noology** — Intellectual machines, logic, tongues, and algebra" and its Japanese original "**Mathematical Psychology** — Thinking machines, logic, languages, and algebra" (hereafter MN and MP, respectively).

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Introduction

I lifted up mine eyes unto the hills, from whence truth hath come.

Anyone should know the non-mathematical part of FE, or at least the spirit of the new intellectual endeavor proposed there under the name Eusophy. It is crucial both for humankind and for individuals — indeed, it may even be called supreme — in that it leads directly to an answer to what I call the Universal Question:

- (1) What should anyone do,

which emerges from one's view of life, and it aims at

- (2) The pursuit of a way for humankind to survive

in view of the following Heating Alarm based on climatology (climate science):

- (3) Global heating was caused by humankind's post-Industrial Revolution way of living, and may become a threat to survival of humankind in the not too distant future if our action is insufficient.

This is supplemented by the following observation:

- (4) The threat posed by global heating recognizes no boundaries.

Therefore, Eusophy transcends various boundaries, both visible and invisible — national, racial, cultural, sectarian, and so on.

Both "survive" in Eusophic Aim (2) and "survival" in Heating Alarm (3) imply continuation as a kind. However, the aim does not entail that the sacrifice of some individuals is permissible. In global crises such as global heating, the survival of humankind as a kind and the survival (continuation of life) of all individuals coincide. This constitutes, in addition to the fact that Eusophy leads directly to an answer to Universal Question (1), another — and perhaps even more important — reason why Eusophy is crucial for individuals as well.

All of the above together strongly urge humankind toward unification in actions that enhance individual life and ensure the survival of humankind. Thus, I have decided to write this Treatise in order to introduce Eusophy to a worldwide general readership.

Not only in the above but also everywhere, I use the term "global heating" in place of the widespread "global warming," in light of the following observation:

- (5) Language is an expression of the mind, yet the mind is also influenced by language.

This interrelationship has both merits and dangers. One danger is that the mind can be misled by inappropriate terms — whether one's own or those of others. "Global warming" may unconsciously evoke the idea of a pleasantly mild climate, leading some to underestimate the severity of the crisis and neglect necessary action.

As its affixes "eu" and "sophy" suggest, Eusophy is, at its core, the pursuit of human omniscience in service of the survival of humankind. It also encompasses a wide range of intellectual endeavors — from theoretical reflection to practical action. Even MP, MN, and FE fall within the scope of Eusophy, for they deepen our understanding of the universe, intellection, and God, and thereby provide part of the foundation for the theoretical reflection undertaken here. The remaining part of the foundation is to be supplied here by deepening our understanding of fundamental aspects of humanity other than intellection, such as emotion, volition, memory, view of life, death, happiness, and moral — and the values underlying all of them.

As stated above, both "survive" in **Eusophic Aim (2)** and "survival" in Heating Alarm (3) imply continuation as a kind, or rather as a species. Therefore, the following basic biological observation is the starting point of Eusophy, the latter part of which I call the Declaration of Humankind's Origin:

- (6) Humans are a kind of organism (such as animals, plants, bacteria, and so on), and humankind is one species among many.

This observation will be emphasized — together with evolutionary theory and climatology — wherever appropriate, for the following three reasons:

1. Even when addressing problems concerning humans, it is helpful to consider their nature as a kind of organism.
2. The law of organic evolution governs how species survive.
3. The mainstay of society still does not seem to be responding adequately to Heating Alarm (3).

On the other hand, although humankind is one species among many, it is not just any species.

- (7) Humankind is distinguished from other species by its intellect and its civilization.

Civilization is defined as the totality of human knowledge acquired through intellect. It therefore serves as the record of humankind's intellectual activity, and through it we can observe both the intellectual evolution of humankind and humankind's most distinctive trait. Indeed, the state of civilization may itself be regarded as a trait of our species — one inherited not through genes, but through language.

Observations (6) and (7) naturally give rise to the idea of two spheres: the biosphere and the humanosphere. The biosphere is the dynamic system composed of all biological communities and their habitats. Its members are interconnected through the biospherical network, which consists of the food cycle (the composite of the food chain and the carbon cycle), symbiotic relationships, and habitat bioprovision — the phenomenon in which part or all of an organism's habitat is provided by other organisms.

In a similar manner, the dynamic system composed of all human communities and their living environments may be called the humanosphere. According to Observation (6), humankind is part of the biosphere, and the biosphere is a system interconnected through the biospherical network. Therefore, the biosphere and the humanosphere overlap to a considerable extent. However, within the humanosphere, non-human organisms and their habitats are regarded as components of the human living environment. Moreover, the humanosphere possesses its own characteristic dynamism and network.

The dynamism and network of the biosphere are exceedingly complex, and much about them remains unresolved. The dynamism and network of the humanosphere, by contrast, should be discernible within the aspects of civilization — the totality of human knowledge, as described above. Although they have long been studied in the humanities, sociology, and history — the disciplines concerned with human beings and human society — they appear to be even more complex than those of the biosphere, and much about them likewise remains unclear. Nevertheless, at least the following observation seems certain.

- (8) Observation about the survival (continuation as a species) of humankind.

- The survival of humankind is inseparable from the survival of its civilization — or more precisely, from the progress of its civilization.
- The sustainability of the biosphere is necessary for the survival of humankind.
- The survival of humankind also requires the progress of the humanosphere. Such progress should include the following:

- The realization of peace and security. Security is a condition that equally guarantees the survival (continuation of life) of diverse individuals and the survival (continuation as a group) of diverse human groups, such as families, ethnic groups, and nations.
- Elevating the United Nations into an international institution whose central mission is the survival of humankind. The United Nations has already begun moving in this direction in practice, through the establishment of organizations such as the WHO (World Health Organization) and, as noted below, the IPCC and the COP. However, as the first official step toward such an elevation, an amendment to the UN Charter is necessary. The Security Council should be transformed into a council dedicated to exploring the path toward a condition in which the survival of humankind is secured.

Observation (8) shows that engaging in activities that promote the sustainability of the biosphere may be regarded as a Eusophic practice. Such engagement must be grounded in biospherology. — the science of the past, present, and future of the biosphere. The biosphere is a dynamic system composed of all biological communities and their habitats, and all its members are interconnected through the biospherical network. Accordingly, the core of biospherology lies in the study of this dynamism and interconnection. Biospherology includes not only the biological disciplines but also the sciences of soil, ocean, and atmospheric systems, for these systems constitute essential parts of the habitats of organisms. It is expected that scholars in these fields will eventually examine the biosphere from the Eusophic point of view, for their disciplines already possess the intellectual foundations required to do so.

Observation (8) also shows that engaging in activities that promote the progress of the humanosphere may be regarded as a Eusophic practice. Such engagement must be grounded in humanospherology. — the study of the past, present, and future of the humanosphere. The humanosphere is a dynamic system composed of all human communities and their living environments, and all its members are interconnected through the humanospherical network. Accordingly, the core of humanospherology lies in the study of this dynamism and interconnection. Humanospherology includes not only the humanities, sociology, and history but also natural science, for the natural world and human world largely overlap. It is expected that scholars in these fields will eventually examine the humanosphere from the Eusophic point of view, for their disciplines already possess the intellectual foundations required to do so.

Such expectations also arise from the following observation concerning humanospheric dynamism and interconnection that is developed in §5.4.5 (**The Venusian Scenario**), which extends the familiar historical claim that "the Black Death opened new opportunities for creativity and wealth, thereby giving rise to Renaissance arts, culture, and concepts, and ultimately to the emergence of modern Europe."

The mid-fourteenth-century pandemic in Europe became a remote cause of the Renaissance, which in turn gave rise to modern European society. As a result, the Industrial Revolution began in Europe, leading to the large-scale use of fossil fuels and the advancement of science and technology. The former initiated massive

CO₂ emissions, and, according to climatologists, global heating has arisen as a consequence. The latter enabled the use of nuclear fission for weaponry and power generation, resulting in concerns about nuclear war and severe accidents at nuclear reactors. Except for the spread of infectious disease, all these developments clearly originate in human agency; yet even the spread of infectious disease ultimately stems from human social life. Thus, these developments are not isolated events but a single continuous chain of humanospheric processes, rooted in human agency and culminating in threats to human survival.

According to **Eusophic Aim (2)**, Eusophy is oriented toward the future of humankind. Therefore, in principle, the emphasis of biospherology and humanospherology should not be placed too far on the past or on regions that are too far removed from the neighbourhood of humankind. However, this principle must be applied with care so as not to dull the insight required to discover close relationships between phenomena that appear, at first glance, to be far apart. This is because, for example, as described above, history and climatology have uncovered the relationship between the Black Death in the fourteenth century and various modern humanospheric events; as described in Observation (10) below, chemists discovered the relationship between the use of CFCs (chlorofluorocarbons) at ground level and the depletion of the ozone layer far above the Earth; and, as explained in §4.1 (**Causality in Science**), Newton discovered the relationship between the falling of objects on Earth and the orbital motion of planets in the heavens — or rather, one of the laws governing the entire universe. Each of these relationships is clearly important for both spherologies.

Observation (8) in particular shows that participating in movements for peace, security, and the elevation of the United Nations may be regarded as a Eusophic practice. Such movements include, for example, the following:

- Efforts to eliminate war and aggression
- Movements to eliminate violence
- Movements to eradicate poverty
- Programs to support the physically vulnerable
- Efforts to improve medical care, public health, and sanitation
- Campaigns to prevent suicide, accidents, and disasters — both natural and human-made
- Efforts to prevent the depletion of essential surface resources — such as freshwater, arable land, fisheries, forests, and habitable land — including efforts to preserve national territories, with particular attention to threats such as submersion and desertification

Although humankind is distinguished from other species by its intellect, as indicated by **Observation (7)**, we must also acknowledge the following. According to the ancient narrative, humankind disobeyed God and ate the fruit of knowledge. This story, though mythological in form, expresses a profound truth about our species:

(9) Humankind is *Homo partim sapiens* — a partially wise being.

In reality, humankind possesses the intelligence to exploit natural resources in countless ways, yet lacks the wisdom to restrain that exploitation. Observations **(11)** through **(14)** below

also support Observation (9). This, I propose, is the true nature of original sin. As a result, we now suffer punishment in the form of global heating that makes life increasingly difficult and threatens the survival of our own species.

(10) Observation about global heating.

- According to **Heating Alarm (3)** and **Supplement (4)**, humankind must confront global heating and transform its way of living with a unified mind.
- To be united in mind means to achieve three forms of shared understanding:
 1. shared recognition of the scientific findings on global heating represented by the IPCC (Intergovernmental Panel on Climate Change) reports;
 2. shared sense of urgency regarding the fact that global heating may become a threat to the survival of humankind in the not too distant future if our action is insufficient; and
 3. shared values that regard the survival of humankind as the supreme good.
- The problem of global heating is strikingly similar to the problem of ozone-layer depletion in three respects:
 1. Both phenomena were discovered by scientists — climatologists in the former case and chemists in the latter — and the scientists centrally involved were awarded the Nobel Prize and are fully recognized within the scientific community.
 2. Both phenomena are global in scale and constitute disasters that threaten the survival of humankind.
 3. Both phenomena are man-made disasters, forms of pollution, caused by massive emissions of gases — emissions of CO₂ in global heating, and emissions of CFCs in ozone-layer depletion. Both such emissions can be stopped if humankind transforms its way of living, because they were driven by the desire for a more comfortable life — or rather, by the unrestrained pursuit of material wealth — and this pursuit was accelerated by the Industrial Revolution.
- In light of the fact that global heating is extremely similar to ozone-layer depletion, an appropriate way of confronting global heating is suggested by the long-standing measures taken against ozone-layer depletion: — scientists conducted further research on ozone-layer depletion; international treaties and domestic laws restricted the use of CFCs; and engineers developed substitutes for them. Following this model, scientists must conduct further research on global heating; international treaties and domestic laws must restrict the use of fossil fuels, which are the main cause of CO₂ emissions; and engineers must develop alternatives to fossil fuels. In short, the collaboration of science, politics, and technology is the core of global heating measures, and the forefront of such collaboration is the COP (Conference of the Parties to the United Nations Framework Convention on Climate Change). However, it seems necessary not only to strive to achieve net-zero CO₂ emissions by 2050, as agreed upon at the COP, but also to implement comprehensive and timely measures before it is too late.
- Transforming humankind's way of living must include overcoming the unrestrained pursuit of material wealth, as well as national egoism — both are causes of global heating and obstacles to effective measures. It must also include ending the excessive economic activities and armed conflicts driven by such pursuit and egoism.

- However, partly due to the nature of humankind indicated by Observations (11) through (14) below, even within the core of society, these three forms of shared understanding remain insufficient. Value conflicts persist in various forms; a lack of urgency is widespread; disregard, skepticism, and neglect toward scientific findings are frequently observed; and, as a natural consequence, effective action for the confrontation and the transformation is stalled.

There is a human characteristic other than (9) that further accelerates global heating and obstructs effective action against it:

- (11) Humankind has evolved a weakened sense of crisis and a tendency toward conflicting value systems.

Possessing the intelligence to foresee even distant death, humans could not have survived if they had lived in constant fear of it. Thus, unless death is imminent, we do not feel its threat — and if this is true of death, it is all the more true of other dangers. Moreover, advanced civilizations provide safety and stability, enabling societies to cultivate diverse value systems that often diverge from the biological imperative of the continuation of life. These evolutionary and civilizational factors together help explain humankind's lack of urgency regarding global heating, and why effective measures have been so slow to advance.

There is yet another human characteristic that further accelerates global heating and obstructs effective action against it. According to **Heating Alarm (3)**, humankind must transform its way of living for the sake of its own future. Yet this transformation is hindered by the following characteristic:

- (12) Humankind finds it difficult to consider the future and, for this reason or other, tends to be deeply attached to the past.

A familiar example illustrates this difficulty. Many science-fiction films that depict the distant future dress their characters in clothing reminiscent of ancient or medieval styles. The deepest attachment to the past is found in religion. Many religions, old or new, are grounded in past authorities such as ancient founders, scriptures, and doctrines, and in this sense they remain strongly attached to the past. A similar pattern is found in political conservatism, which often grounds its legitimacy in historical traditions, established institutions, and inherited values. In this sense, conservatism also remains strongly attached to the past. In society at large, various forms of attachment to the past derive from authoritarianism.

Global heating is a near-future threat to the survival of humankind caused by the way of living adopted over the past several centuries. If one possesses sufficient knowledge and a sufficient sense of urgency regarding global heating, one naturally becomes not only committed to global heating measures, but also future-oriented and reform-minded, in the sense that one recognizes the necessity of transforming the way of living that has prevailed until now. However, in politics, the boundary between emphasis and neglect of global heating measures does not coincide with the boundary between reformism and conservatism. In fact, Japan's reformist political parties, though presumably future-oriented, inexplicably tend to place little emphasis on global heating measures. The current ruling conservative party has declared, in accordance with the COP, that it will strive to achieve net-zero CO₂ emissions by 2050. Yet in some countries, these boundaries are mistakenly assumed to coincide, leading

conservative parties to attack global heating measures in order to attack reformist parties, and leading reformist parties to attack the neglect of global heating measures in order to attack conservative parties. In other words, there is a tendency to use global heating as a tool for political conflict. As noted in Observation (10), the collaboration of science, politics, and technology is the core of global heating measures, and since political divisions between conservatism and reformism exist in many countries, this tendency is extremely serious. In countries where religion has a strong influence on politics, the attachment to the past found in religion is equally serious. After all, as stated in Observation (10), humankind must confront global heating and transform its way of living with a unified mind.

Even in academic fields that value new knowledge — which necessarily implies criticism of past authorities — attachment to the past is not uncommon: history, archaeology, paleontology, and the sciences that investigate the origins of the universe, Earth, life, and humankind all focus on the past; so do disciplines that analyze past thought. Mathematics, though free from attachment to the past, is equally free from orientation toward the future, enabling mathematicians to dwell in a timeless universe. As will be discussed in §5.4.2 (**Essence of Intellectual Exploration**), some of the attachment to the past found in academic fields — and the mathematician's attitude of being indifferent to both past and future — seem to derive from **Interrelationship (5)**.

There is yet another human characteristic that further accelerates global heating and obstructs effective action against it:

(13) Humankind pursues material wealth without limit.

This tendency, greatly amplified by the Industrial Revolution, has been a major driver of global heating. According to Heating Alarm (3), humankind must transform its way of living for the sake of its own future. The core of this transformation lies in overcoming this unrestrained pursuit of material wealth. Yet both in the international community and within individual societies, material wealth is limited in total quantity and is unevenly concentrated among wealthy nations and wealthy classes. Those in other nations or social strata, who must raise and maintain their quality of life to a reasonable level, will inevitably need to continue pursuing material wealth for some time to come. Therefore, it is essential for the international community and for each society to establish mechanisms for redistributing wealth — mechanisms that reduce disparities in the ownership of material wealth between nations and among citizens — in order to shorten this period of pursuing material wealth as far as possible. And the difficulty of establishing such mechanisms in the international community is likely the principal reason why discussions at the COP often become contentious.

There is yet another human characteristic that further accelerates global heating and obstructs effective action against it:

(14) Humankind is prone to conflicts driven by partial love, such as patriotism and sectarianism.

This tendency has repeatedly culminated in armed conflicts, as seen in many regions of the world even today. According to Heating Alarm (3) and Supplement (4), humankind must unite in order to confront global heating. Yet partial love stands in direct opposition to such unity, and the use and production of weapons further accelerates global heating.

Although humankind is one species among many, as stated in Observation (6), it is not just any species. Unlike other species, humans have sublimated the self-love and familial love inherent in the biological survival instinct into broader forms of love — such as tribal, ethnic, and national love. This sublimation was surely an important part of the process by which our species evolved from beast to human — *Homo partim sapiens*. Further evolution will involve elevating these forms of partial love into love for humankind as a whole — a transformation crucial for confronting global heating. In this sense, global heating, though an unfortunate development, also provides humankind with a rare opportunity for further evolution — a step toward true *Homo sapiens*.

Human mentality and conduct are guided toward the high or the low depending on whether one possesses sufficient alarm about global heating. Figuratively speaking, global heating stands as a signpost placed at that point of bifurcation. Clarifying the path toward the high amounts to clarifying — in the context of Eusophy — the relationships among the following seven questions, the first of which is **Universal Question (1)**. The answers to these questions may vary widely depending on one's values. Therefore, discussing them without first establishing the guiding value system would be fruitless. In this Treatise, they are examined under the value system inherent in Eusophy. Under such guidance, the relationships among these questions can be read almost effortlessly and with firm conviction.

1. What should anyone do?
2. What is the meaning of life?
3. What is the supreme good?
4. What is happiness?
5. What is humanity?
6. What is life?
7. How should one view life and death?

Suppose one starts from **Observation (6)**: "Humans are a kind of organism, and humankind is one species among many." If one moreover sincerely accepts **Heating Alarm (3)** and **Supplement (4)**, a sufficient sense of crisis concerning the survival of humankind will naturally arise, and one will naturally arrive at the following:

Answer to Question 3: The supreme good is the survival of humankind,

since the survival of humankind constitutes the greatest good for the greatest number — namely, the entire species, from the past to the future. This answer together with **Eusophic Aim (2)**: "The pursuit of a way for humankind to survive as a species" suggest that Eusophy may itself be called supreme. If one moreover is contemplative or action-oriented, one will naturally be led further to theoretical reflection or practical action toward Eusophic Aim (2), and thereby participate in Eusophy.

The proposal and the supremacy of Eusophy together make many previous activities fade in comparison and seriously pose Question 1. Eusophy, however, also presents many alternative activities — glorious and attainable — and the most humane answer to Question 1.

What, then, are the alternative activities that are glorious and attainable? They will naturally be found abundantly in the explanations of **Observations (8)** and in **Observation (10)**: global heating measures, biospherology, humanospherology, realizing peace and security, elevating the United Nations into an international institution whose central mission is the survival of humankind, and so on.

What, then, is the most humane answer to Question 1? Since one has started from Observation (6), one will naturally ask Question 6. The proper answer to it will be presented in §2 (**Life and the Biosphere**). From there, one will naturally be led to the following view of life and death:

Answer to Question 7: Figuratively speaking, each individual life is one baton in a relay without goal or victory; each of us is given the important role of receiving and passing the baton as a member of some team; and if one can leave the stage with the assurance that one has fulfilled that role, such a departure cannot be called unhappy.

If one moreover believes in human dignity, one will naturally ask "What distinguishes humankind from other species?" and will naturally arrive at the following:

Answer to Question 5: Humankind is a species distinguished from other species by its intellect and its civilization,

where civilization is defined as the totality of human knowledge acquired through intellect. This answer is **Observation (7)**. In light of these two answers, one will naturally arrive at the following view of life:

Answer to Question 2: Life becomes meaningful if one receives and passes on the baton of life and civilization.

And then one will naturally arrive at the following view of happiness:

Answer to Question 4: The happiness appropriate for human beings is the happiness obtained by living in accordance with the view of life given by Answer to Question 2.

Finally, one will naturally arrive at the following:

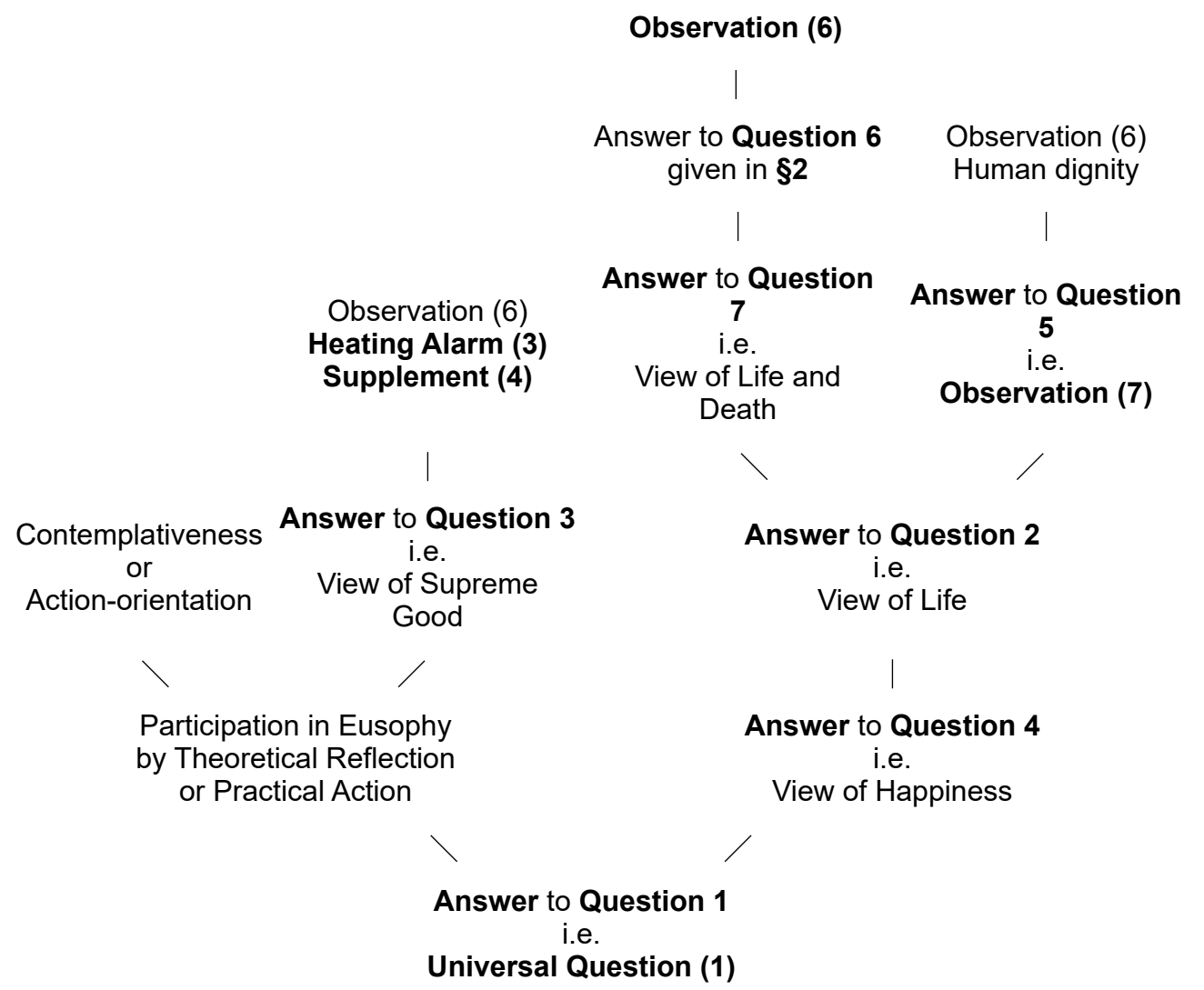
Answer to Question 1: What anyone should do is to participate in Eusophy and pursue the happiness described in Answer to Question 4, if possible.

This is the most humane answer to Question 1.

The frequent use of "naturally" in these explanations reflects the fact that the "arrivals" described here came to me almost effortlessly and with firm conviction. This fact is remarkable, considering that Question 3 has been pondered since ancient Greece and Rome, that an intellectual endeavor like Eusophy — undeniably urgent for humankind — has never before been proposed, and that people, even when they ask these questions, are likely wandering in confusion. That I arrived at Answers to Questions 1 through 7 — without effort and with firm conviction — is precisely because I have been walking in the right direction at

the fork where the signpost of global heating stands. Age is no obstacle to such progress. Until my mid-sixties, I wandered far before that signpost, without ever asking these questions. Therefore, I write this Treatise in the hope that many others will, even now, follow this correct path.

Eusophic Diagram



Terminologies

This section explains the differences between everyday terminology and the terminology used in this Treatise, which necessarily engages with academic concepts and therefore requires a somewhat more precise handling of ideas.

Contents

- 1. Civilization
- 2. Action
- 3. Culture

Terminology 1. Civilization

As already stated in the Introduction, civilization is defined as the totality of human knowledge acquired through intellect. This usage of "civilization" is not only appropriate for the purposes of this Treatise, but is also reasonable, for "civilize" includes the meaning "to bring out of an uneducated state," and "educated" naturally implies "possessing a certain amount of knowledge."

Terminology 2. Action

In this Treatise, "action" is used as the most general term for "human physical events." In other words, an action is an intellectual object that occurs as a relatively direct result of the activity of human effectors — that is, muscles and sensory organs. This explanation follows the definition "An intellectual object is called an event if it makes sense to ask whether it occurs" given in §1.2 (**Answer to Question 2**). Actions in this broader sense include not only actions in the ordinary sense, but also acts, deeds, conducts, behaviors, and so on.

Incidentally, Japanese and Chinese already possess words corresponding to this meaning of "action," such as 「行ない」 and 「行為」. Since English appears to lack terms that correspond to these, the above terminology becomes necessary.

Terminology 3. Culture

In everyday life, "culture" is often used ambiguously and is sometimes even confused with "civilization." In this Treatise, however, culture is defined as the totality of pan-social behavioral modes — that is, behavioral modes shared by the great majority of members of a society. In spite of the above broader notion of "action" including behavior, behaviors here are interpreted as including not only physical ones but also mental ones. Under this interpretation, the culture of a society can be divided into two parts: mental culture and physical culture. Mental culture consists of the pan-social mental behavioral modes. Physical culture consists of the pan-social physical behavioral modes — that is, the pan-social modes of using the human body, matters, and techniques.

The so-called "national character" may be understood as the mental culture of a national society. The various views that concern this Treatise — such as the views of the cosmos, universe, God, world, humanity, life, death, and happiness — are modes of viewing objects, and therefore belong to the mental culture of a society, provided that such modes are pan-social. The various senses that concern this Treatise — such as senses of value, crisis, moral, and beauty — likewise belong to the mental culture of a society, for the same reason and under the same condition. Sportsmanship also belongs to mental culture, for it is the sense moral concerning the actions within play.

The mode of using language belongs to culture. Language is originally expressed by using the human body — that is, by vocal expressions, facial expressions, and gestures. More generally, it is expressed by using matters, that is, symbols including phonemes, bodily figures, letters, and Braille. It is also expressed by using techniques of selecting symbols to use and arranging them. The mode of expressing language is of course pan-social, and thus belongs to physical culture. By contrast, the mode of understanding language belongs to mental culture.

The mode of living concerns culture. To see this, let us consider primitive human societies, for modern living, in its essence, is the same as the primitive one. Primitive humans lived by hunting and gathering in groups for food, by sewing pelts for clothing, and by combining natural things for shelters. The mode of using the human body, matters, and techniques for these actions may be called the mode of survival strategy, and belongs to physical culture. Primitive humans also thought that order in societies was necessary in order to maintain or improve their society. Such order is now called convention, custom, tradition, and so on, and the mode of viewing these belongs to mental culture.

For obvious reasons, the modes of science, logic, ethics, and religion as discussed in §4 (**Causality in Science, Logic, Ethics, and Religion**) and §5.1 (**The Common Origin of Science, Logic, Ethics, and Religion**) belong to mental culture. It will be shown in §4 that they are included in the mode of understanding causality.

The Two Aspects of Culture

Aspect	Mental Culture	Physical Culture
Description	The pan-social mental behavioral mode	The pan-social physical behavioral mode
Examples	The national character Pan-social views and senses The mode of understanding language The mode of viewing order in societies The mode of understanding causality	The mode of expressing language The mode of survival strategy

As explained above, culture and civilization are unmistakably different, yet they are related. For example, the pan-social modes of acquiring knowledge constituting civilization belong to the mental culture of that society. Conversely, every culture involves the knowledge constituting civilization or the products of civilization.

Like civilization, culture also has its own products. For example, scholarship and arts are products of culture — specifically, of modes of intellection and of the sense of beauty. Similarly, just as civilization and culture have products, so do the various views. Notions are among their principal products. For example, the notion of God relevant to §1 (**The Universe, Intellection, and God**) is a product of the view of God — that is, of the mode of considering God.

Since culture is defined as the totality of pan-social behavioral modes — that is, behavioral modes shared by the great majority of members of a society — each individual's behavioral modes, while largely aligned with the culture of the society to which they belong, also contain aspects that do not align with it, namely individuality. Like culture itself, individuality has both mental and physical aspects. The mental aspect is what is commonly called "character," while the physical aspect is what is commonly called "habit."

Incidentally, the so-called "humanities" may be understood as the academic disciplines that study culture and the products of culture.

1. The Universe, Intellection, and God

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1.1. Three Questions

Although I do not adhere to any established religion, I sometimes find myself drawn to religious spaces. At a Christmas Eve service in 2012, I encountered a verse from the "Gospel of John" to the following effect:

All things were made by the Word.

This verse astonished me, for reasons I explain below.

Since 1992, I have been studying intellect (HI, human intelligence) in **MP** and **MN** through mathematical science — science by means of mathematical models — to be discussed in §A.3 (**Mathematical Science and Models**). This inquiry was motivated by the question "What is humanity?" and guided by Observations **(6)** and **(7)**. Intellect is inseparably linked to language, in the following sense:

(IL) Intellect evolved because humankind acquired language.

This evolutionary process may be outlined as follows:

1. In early human societies, intellect existed prior to language.
2. Language first emerged as a tool for communication with others.
3. It then became a tool for internal communication.
4. Consequently, it developed into a medium for thought and memory.
5. It then became a tool for transmitting knowledge and ways of thinking to future generations.
6. Consequently, it became a medium through which intellect itself is inherited.
7. Thus, intellect evolved.

This entire process may be encapsulated in **the above phrase**: "All things were made by the Word." Therefore, I was astonished, because it seemed to me that John had already known everything I had just described.

The preaching, however, was disappointing. According to it, "Word" in the "Gospel of John" does not mean language, but means God's words such as "Let there be light" for creating the heaven and the earth. With some hope, however, I turned to the World Wide Web, found that the "Wisdom of Solomon" has something to do with "Word," and found out the following two verses in it:

As for wisdom, what she is, and how she came up, I will tell you, and will not hide mysteries from you: but will seek her out from the beginning of her nativity, and bring the knowledge of her into light, and will not pass over the truth.

God hath granted me to speak as I would, and to conceive as is meet for the things that are given me: because it is he that leadeth unto wisdom, and directeth the wise.

These verses seemed mostly adequate to a prologue and an epilogue, respectively, of a certain section of MN which amplifies **Observation (IL)**. The latter, however, also seemed inadequate to MN, because I had long felt that science was incompatible with God, while noology in MN is the science of intellectual phenomena.

The above inadequacy, however, marked the beginning of a fortunate chain of unexpected insights. It first led me to the following question:

Question 1: What is God?

And since they say that God is the omnipotent ruler of the universe, this prompted me to consider the following further questions:

Question 2: What is the universe?

Question 3: What is the omnipotent rule over the universe?

Moreover, it soon occurred to me that these questions, as well as intellect, could be addressed elegantly by the mathematical models constructed in MP and MN.

1.2. Answer to Question 2

Intellection is the activity of knowing the objective world, and intellect is the capacity (the actual ability and the potential ability) to know that world. Accordingly, the objective world is the totality of intellectual objects.

An intellectual object is called an entity if it makes sense to ask whether it exists. For instance, "UFOs" and "Peter" (in "The Tail of Peter Rabbit") are entities. An intellectual object is called an event if it makes sense to ask whether it occurs. For instance, "UFOs are round" and "Peter eats a radish" are events, as well as their parts "are round," "eats a radish," and "eats." Entities may informally be referred to as things and, together with events, as matters.

Not all but many intellectual objects may be expressed by language. Linguistic expressions referring to entities are called nominals, while those referring to events are called declaratives.

Observation of language and introspection show that the intellectual objects are divided into entities and events. Observation of language also shows that each event is either an attribute or an act of an entity (and the attribute or the act itself) or a relation among entities (and the relation itself). They are referred to as an attribute event, an act event, or a relation event, respectively.

For instance, the event "UFOs are round" is an attribute of the entity "UFOs" and the event "are round" is the attribute itself. The event "Peter eats a radish" is a relation between the entities "Peter" and "a radish," and the event "eats" is the relation itself. The event "Peter eats a radish" is also an act of the entity "Peter" and the event "eats a radish" is the act itself, which is also an attribute of the entity "a radish," showing that "a radish" is eaten by someone, and the event "eats" is also the attribute itself.

The objective world is vast. People engage in intellection only within limited ranges of intellectual objects, according to their interests. The objective world, then, consists of such limited ranges. Because of **Observation (IL)**, we can observe these ranges through language. Moreover, in the course of mathematical science in **MP** and **MN**, I have been able to construct mathematical models of such ranges, on the basis of that observation. In terms of these models, the limited ranges are called the denotable worlds. Here, I call them the universes. This is my answer to **Question 2**: "What is the universe?" There are various universes, and the objective world is a multiverse composed of these universes.

In each universe, an entity x exists if and only if the event " x exists" occurs. Conversely, an event X occurs if and only if the entity "that X occurs" exists. In this way, the pairs (entity, existence) and (event, occurrence) correspond to each other. We shall soon see that causality governs the occurrence of events, and thereby governs the existence of entities through the above correspondence. In this sense, the objective world is an orderly system. Moreover, since it is the totality of intellectual objects, it was created by our intellect. This perspective stands in stark contrast to religious belief, in which the universe is typically regarded as the creation of a transcendent deity. It also stands in stark contrast to prevailing physical cosmology, in which the universe is understood to have originated with the Big Bang.

Given this, the objective world and the universes should be called the noo-cosmos and noo-worlds, respectively, and an intellectual object in a noo-world should be called a noo-element of the noo-world — "noo-" and "cosmos" derive from Greek "nóos" and "kosmos," meaning "intellect" and "order," respectively.

Thus, the noo-cosmos is composed of noo-worlds, whose noo-elements are entities and events. In the mathematical science mentioned above, the noo-worlds are defined as certain sets, whereas the noo-cosmos cannot be defined as a set due to a certain set-theoretical reason to be clarified in §A.1 (**Noological Set Theory**). Therefore, we use the set-theoretical term "element" for noo-worlds, but not for the noo-cosmos.

1.3. Causality and Necessity

As stated above, in each universe, the pairs (entity, existence) and (event, occurrence) correspond to each other. In other words, these pairs are equally essential for the universe. However, we shall soon see that, for human intelligence, the true essence lies in the pair

(event, occurrence) — or more precisely, in causality, that is, the totality of causal connections among events.

Let $A_1, \dots, A_n$, and B denote events, where n is a nonnegative integer. Then the following six expressions are equivalent by definition:

1. If all of $A_1, \dots, A_n$ occur, then B occurs.
2. The tuple $(A_1, \dots, A_n)$ implies B , that is, $A_1, \dots, A_n$ together imply B .
3. $A_1, \dots, A_n$ together are causally connected to B .
4. B is an effect of the causes $A_1, \dots, A_n$.
5. A_1 implies B under $A_2, \dots, A_n$.
6. $A_1, \dots, A_n \Rightarrow B$.

If $n = 0$, then $A_1, \dots, A_n$ should be understood not to exist, and similarly for $A_2, \dots, A_n$ in the case $n = 1$. This definition in particular says that the symbolic expression 6 may be read as the expressions 1 through 5, especially as 2 or 3.

If $n = 0$ in the expression $A_1, \dots, A_n \Rightarrow B$, it is denoted by $() \Rightarrow B$. It means that B necessarily occurs, that is, $C_1, \dots, C_m \Rightarrow B$ for every tuple $(C_1, \dots, C_m)$ of events. This is because, if all of $C_1, \dots, C_m$ occur, then, as $n = 0$, all of $A_1, \dots, A_n$ obviously occur, and so B occurs. Therefore, if $() \Rightarrow B$, B is also said to be necessary. For instance, the event " x is x " is necessary for any entity x . A law is an event which is supposed to be necessary. We may replace "necessary" by "inevitable" and "unavoidable." In religious context, we may say that B has been determined by Providence (God's will). This naturally leads to an answer to the question "What is Providence?" to be presented in §1.5 (**Answer to Questions 1 and 3, and Much More**).

Let B be an event. If B is not necessary, we naturally say that B is unnecessary, evitable, or avoidable. If the negation $\neg B$ of B is necessary, we say that B is impossible. Then, since $\neg(\neg B)$ is equivalent to B , $\neg B$ is impossible if and only if B is necessary. For instance, the event " x is not x " is impossible for any entity x . If B is not impossible, we naturally say that B is possible. If B is neither necessary nor impossible, that is, if B is unnecessary and possible, we say that B is contingent. It is a good logical exercise to derive the following:

- B is possible if and only if $\neg B$ is unnecessary.
- B is unnecessary if and only if $\neg B$ is possible.
- B is contingent if and only if $\neg B$ is contingent.

B	necessary	impossible	possible	unnecessary	contingent
$\neg B$	impossible	necessary	unnecessary	possible	contingent

As will be shown by **Proposition on Paradoxes** in §A.2 (**Mistakes in Paradoxes**), no event is both necessary and impossible. Therefore, every event is either unnecessary or possible. In other words, necessary events are possible, and impossible events are unnecessary. For instance, for any entity x , the event " x is x " is possible, and the event " x is not x " is

unnecessary, or rather, the proof of the above proposition is based on the fact that "x is x" is possible.

necessary	unnecessary	
	contingent	
possible		impossible

Examples on Causality and Necessity

Example 1: Let x denote an arbitrary entity and let A and B denote the event " x is a planet" and the event " x orbits the sun," respectively. Then Newton has shown that A implies B under the law of universal gravitation, the laws of motion, and certain implicit laws, such as the laws of equality between intellectual objects.

Example 2: Let A be the event "Humankind has emitted too much CO_2 (carbon dioxide) since the Industrial Revolution," B be the event "The global temperature is rising," and C be the action of achieving net-zero CO_2 emissions by 2050. Then climatologists assert, in support of **Heating Alarm (3)**, that $\neg A \Rightarrow \neg B$ and $C \Rightarrow \neg A$, hence $C \Rightarrow \neg B$, that is, the action C can stop the event B after 2050. On the other hand, some people assert that no action can stop B because B is necessary, or in religious context, B has been determined by Providence.

Newtonian mechanics was initially viewed with skepticism, even within the academic world. Yet as it came to be recognized that its predictions accorded well with observational facts, it gradually gained acceptance. Today, even among the general public, most people believe in Newtonian mechanics despite being unable to understand its details.

Modern climatology followed a similar trajectory: it too was once met with skepticism, even within academia, but as its findings came to align closely with observational facts, it likewise gained credibility. However, in contemporary society, there appear to be many people — including those at the core of society — who not only fail to understand the details of climatology but also refuse to believe it at all. At what point, then, will the central sectors of society come to accept climatology? Will they refuse to believe it until undeniable observational facts occur — that is, until the heat surpasses tolerable limits and ordinary life becomes impossible?

In this connection, one is reminded of the case of ozone-layer depletion. The central sectors of society accepted the chemists' finding — that CFCs emitted by human activity destroy the ozone layer, thereby increasing ultraviolet radiation and threatening the survival of humankind — even without understanding its details, and immediately acted in unison to implement measures.

This chemical finding is analogous in form to the climatologists' finding that CO_2 emitted by human activity traps solar heat, causing global temperatures to rise and thereby threatening the survival of humankind. Yet despite this analogy, one was readily believed, while the other is persistently refused belief.

The reason for this contrast is, to me, easy to imagine — and likely so for many others as well. And toward that reason, I cannot help but feel a profound sense of gloom.

1.4. Omnipotence and Omniscience

Observation of language shows that there exist certain operations on the events. Each operation assigns an event to each tuple $(A_1, \dots, A_n)$ of events, for a positive integer n called the arity of the operation. If the operation is denoted by F , then the event which F assigns to $(A_1, \dots, A_n)$ is denoted by $F(A_1, \dots, A_n)$.

For each event A , $\neg A$ denotes the negation of A . The symbol " $\neg$ " stands for the expression "It is not the case that" and therefore is prepositive (incidentally, it should be postpositive in the Japanese word order). If A denotes, say, the event "UFOs are round," then $\neg A$ is the event "It is not the case that UFOs are round," which is equivalent to the event "UFOs are not round." For each pair (A, B) of events, $A \& B$ denotes the conjunction of A and B . For instance, if A and B denote the events "Peter eats a radish" and "Peter eats a lettuce," respectively, then $A \& B$ is the event "Peter eats a radish and Peter eats a lettuce," which is equivalent to the event "Peter eats a radish and a lettuce."

The conjunction operation assigns each pair (A, B) of events their conjunction $A \& B$. It is a binary operation and denoted by the symbol $\&$, hence $\&(A, B) = A \& B$. The negation operation assigns each event A its negation $\neg A$. It is a unary operation and denoted by the symbol " $\neg$," hence $\neg(A) = \neg A$. The negation operation and the conjunction operation are basic ones. There exists a variety of operations, because operations can be composed. For instance, composing the negation operation and the conjunction operation, we obtain the operation which assigns each pair (A, B) of events the event $\neg(\neg A \& \neg B)$, although it is equivalent to the basic event " $A \text{ OR } B$."

Let F be an n -ary operation on the events for a positive integer n . Then F is called a deduction if $A_1, \dots, A_n \Rightarrow F(A_1, \dots, A_n)$ for each tuple $(A_1, \dots, A_n)$ of events. For instance, the conjunction operation $\&$ is a deduction, while the negation operation " $\neg$ " is not a deduction, because " x is x " does not imply " $\neg(x \text{ is } x)$ ", or more generally, no necessary event A satisfies $A \Rightarrow \neg A$, as will be shown by **Law 8** and **Proposition on Paradoxes** in §A.2 (**Mistakes in Paradoxes**).

Let F be an n -ary operation on the events for a positive integer n , and C be a collection of events. Then $F(C)$ denotes the collection of all events $F(A_1, \dots, A_n)$ for all tuples $(A_1, \dots, A_n)$ of events in C . This is different from mathematical notation, where $F(C)$ is denoted by $F(C^n)$.

Let C be a collection of events, and B be an event. If, for some nonnegative integer n , there exists a sequence $A_0, \dots, A_n$ of events satisfying the following two conditions, then we say that B is generated by C and call the sequence an explanation of B by C :

- $A_n = B$
- For each integer k from 0 through n , the event A_k satisfies one of the following two conditions:
 1. A_k belongs to C .
 2. A_k belongs to $F_k(\{A_0, \dots, A_{k-1}\})$ for some deduction F_k .

In the condition 2, $\{A_0, \dots, A_{k-1}\}$ denotes the collection of $A_0, \dots, A_{k-1}$. As before, if $k = 0$, then $A_0, \dots, A_{k-1}$ should be understood not to exist. Therefore, A_0 does not satisfy the condition 2, and so it belongs to C.

Obviously, the following holds:

Proposition on Generation and Explanation

Let C be a collection of events, and B be an event. Then B is generated by C if and only if there exists an explanation of B by C. If B is generated by C, then it is also generated by every collection of events containing C.

Let $A_0, \dots, A_n$ be an explanation of B by C. Then, for each integer k from 0 through n, the sequence $A_0, \dots, A_k$ is an explanation of A_k by C. Suppose A_k satisfies the condition 2 above and k' is the arity of the deduction F_k in it. Then there exists a tuple $(B_1, \dots, B_{k'})$ of events in $\{A_0, \dots, A_{k-1}\}$ for which $A_k = F_k(B_1, \dots, B_{k'})$, and therefore $B_1, \dots, B_{k'} \Rightarrow A_k$, because F_k is a deduction. Since $B_1, \dots, B_{k'}$ belong to $\{A_0, \dots, A_{k-1}\}$, it follows that $A_0, \dots, A_{k-1} \Rightarrow A_k$. It is a good logical exercise to apply these facts to an inductive proof of the following:

Proposition on Generation and Causality

Let C be a collection of events, B be an event, and assume that B is generated by C. Then there exist events $C_1, \dots, C_m$ in C for which $C_1, \dots, C_m \Rightarrow B$.

Let L be a collection of laws. If L has the following property called omnipotence, then we say that L is omnipotent:

The following two expressions are equivalent for each tuple $(A_1, \dots, A_n, B)$ of events:

1. B is an effect of the causes $A_1, \dots, A_n$.
2. B is generated by the union $L \cup \{A_1, \dots, A_n\}$.

Here, $\{A_1, \dots, A_n\}$ denotes the collection of $A_1, \dots, A_n$, and $L \cup \{A_1, \dots, A_n\}$ is the union of the collections L and $\{A_1, \dots, A_n\}$. It follows from Proposition on Generation and Causality that condition 2 implies condition 1. Therefore, the omnipotence of L, in fact, means that, for each tuple $(A_1, \dots, A_n, B)$ of events, condition 1 implies condition 2. A similar remark applies to the following definition of omniscience.

If L has the following property called omniscience, then we say that L is omniscient:

The following two expressions are equivalent for each tuple $(A_1, \dots, A_n, B)$ of events:

1. B is an effect of the causes $A_1, \dots, A_n$.
2. There exists an explanation of B by $L \cup \{A_1, \dots, A_n\}$.

Then Proposition on Generation and Explanation shows that the following proposition holds: